ModelIt!

Teacher's Guide

The Invisible Plant Factory

How Plants Build Themselves from Thin Air and Sunlight

Quick Reference

Grade Level	Grade 6
NGSS Standard	MS-LS1-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS1-6

Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model showing how sunlight, water, CO_2, and chlorophyll interact to drive photosynthesis and produce glucose.
Disciplinary Core Idea	**LS1.C Organization for Matter and Energy Flow in Organisms** — Plants use the energy from light to make sugars from carbon dioxide and water through photosynthesis, which releases oxygen. This is the entry point for energy and carbon into living systems.
Crosscutting Concept	**Energy and Matter: Flows, Cycles, and Conservation** — Students trace the flow of energy (sunlight → chemical energy in glucose) and matter ($CO_2 + H_2O \rightarrow C_6H_{12}O_6 + O_2$) through the photosynthesis process.

CAST Testing Alignment

MULTIPLE CHOICE:

A student grows two identical plants — one in sunlight and one in complete darkness. After 4 weeks, the plant in sunlight is thriving but the one in darkness has wilted. Which best explains this result?

CONSTRUCTED RESPONSE:

A scientist weighs a potted tree seedling and its soil. After 5 years, the tree has gained 50 kg but the soil has only lost 0.5 kg. Explain where the other 49.5 kg of mass came from, using the process of photosynthesis.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Photosynthesis is the most important chemical reaction on Earth. It converts solar energy into food energy, produces the oxygen we breathe, and is the foundation of almost every food chain on the planet.

The Photosynthesis Equation

$6CO_2 + 6H_2O$ + light energy → $C_6H_{12}O_6 + 6O_2$. In words: six molecules of carbon dioxide plus six molecules of water, powered by light energy, produce one molecule of glucose and six molecules of oxygen. This seemingly simple reaction involves hundreds of molecular steps and is the basis of almost all life on Earth.

Where Does the Mass Come From?

This is the most counterintuitive fact in plant biology: most of a tree's mass comes from CO_2 in the air, NOT from the soil. The famous Van Helmont experiment (1648) showed that a willow tree gained 75 kg while its soil lost only a few ounces. The carbon in CO_2 is rearranged into organic molecules that build wood, bark, leaves, and roots.

Limiting Factors

Photosynthesis rate is limited by whichever input is in shortest supply — the 'law of the minimum.' On a sunny day with plenty of water, CO_2 is usually the limiting factor. On a cloudy day, light is limiting. During drought, water is limiting. This concept is critical for agriculture — farmers must provide ALL needed inputs for maximum crop yield.

Global Importance

Plants and algae perform photosynthesis, producing about 300 billion tons of oxygen per year. Photosynthesis is also Earth's main carbon sink — removing CO_2 from the atmosphere. Deforestation reduces this capacity, contributing to climate change. Understanding photosynthesis is essential for addressing food security and climate challenges.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify sunlight intensity, water availability, CO_2 concentration, and chlorophyll amount as the key inputs to photosynthesis.

E **Establish Relationships**
Students determine that all three external inputs positively affect photosynthesis rate (more input = more glucose), and that chlorophyll amount determines how much light the plant can capture.

V **Visualize & Model**
Students build a model showing how limiting factors control glucose production and how removing any single input stops the entire process.

E **Evaluate Outcomes**
Students run perfect day, cloudy, and drought scenarios to identify which factor limits photosynthesis under different conditions.

R **Revise & Extend**
Students add temperature or leaf surface area to explore how additional factors affect photosynthesis in real plants.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"A tree weighing 10,000 pounds grows from a tiny seed. Where does all that STUFF come from?"*

Watch For

Hold up a small seed and a photo of a giant tree. Ask: Where did the mass come from?

Slide 2: Learning Objectives

3 min

Say: *"Today we're cracking the mystery of how plants build themselves from thin air."*

Watch For

Have students read objectives. Pre-teach 'photosynthesis,' 'chlorophyll,' and 'glucose.'

Slide 3: Big Question

3 min

Say: *"Here's a mind-blowing fact: most of this tree is made from AIR. Not soil. Not water. Air."*

Watch For

Poll the class: Soil, water, air, or sunlight? Most students will say soil — they're wrong!

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model of the invisible factory inside every leaf."*

Watch For

Preview LEVER steps. Connect to prior knowledge — what do students know about how plants grow?

Slide 5: Activity 1: Components

8 min

Say: *"What does a plant need to make food? Which inputs come from the environment?"*

Watch For

Guide sorting. Emphasize that THREE inputs are all external — plants depend entirely on their environment.

Slide 6: Activity 2: Connections

8 min

Say: *"Every input is positive — more sunlight, more water, more CO_2 all mean more food. But remove ANY one and it all stops."*

Watch For

Demonstrate with a recipe analogy: you can't make a cake without flour, even if you have eggs and sugar.

Slide 7: Activity 3: Simulation

10 min

Say: *"What happens to a plant on a cloudy week? What about during a drought?"*

Watch For

Students predict, then run scenarios. Compare which limiting factor has the biggest impact.

Slide 8: Discoveries

5 min

Say: *"Trees are made from AIR. Your desk was once CO_2 floating around in the atmosphere."*

Watch For

Reveal the Van Helmont experiment result. Connect to carbon cycle — where does the carbon go?

Slide 9: STEM Challenge

12 min

Say: *"NASA needs your help. Design a greenhouse that works on MARS!"*

Watch For

Distribute Mars environment fact sheets. Groups design greenhouses optimized for Martian conditions.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Sunlight Intensity, Water Availability, CO_2 Concentration
- **INTERNAL:** Chlorophyll Amount

Relationships

- **Sunlight Intensity to Glucose Production:** POSITIVE (+) — More sunlight provides more energy to drive the photosynthesis reaction. Up to a saturation point, increased light directly increases glucose production.
- **Water Availability to Glucose Production:** POSITIVE (+) — Water is a direct reactant in photosynthesis ($6H_2O$ needed per glucose molecule). Without water, the reaction cannot proceed regardless of other inputs.
- **CO_2 Concentration to Glucose Production:** POSITIVE (+) — CO_2 provides the carbon atoms that become glucose. More CO_2 allows the plant to build more sugar molecules, up to the point where another factor becomes limiting.

Key Discoveries

- Plants need ALL THREE inputs (sunlight, water, CO_2) — removing any one stops photosynthesis completely
- Most of a tree's mass comes from CARBON DIOXIDE in the air, not from the soil
- Photosynthesis is the reverse of cellular respiration — they're balanced opposite reactions
- More chlorophyll allows a plant to capture more light energy, increasing photosynthesis rate up to a saturation point

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Plants get their food from soil"	Plants get WATER and MINERALS from soil, but their actual food (glucose) is made through photosynthesis using CO_2 from air and light energy. Soil provides micronutrients but is NOT the main source of plant mass. Van Helmont proved this in 1648.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Plants only do photosynthesis, not respiration"	Plants do BOTH. During the day, photosynthesis produces more oxygen than respiration consumes, so the net effect is oxygen release. At night, plants only respire (consuming oxygen, releasing CO_2) just like animals. Plants are full-time respirers and part-time photosynthesizers.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Photosynthesis is the reverse of breathing"	While the equations are reverse, the processes are completely different. Photosynthesis uses light energy to BUILD glucose molecules (energy storage). Respiration BREAKS glucose to release energy (energy use). They happen in different organelles (chloroplasts vs. mitochondria) using different enzymes.